

Project 1 - The Grader

CS496 AI Engineering - Final Report

1. Task and why exact match is not enough

This project evaluates customer-support replies generated from a short policy and a customer message. There is no single correct wording for many support questions: different responses may communicate the same valid decision in different ways. Therefore, exact string matching would incorrectly reject many good responses.

We evaluate each reply using four dimensions:

Correctness - whether the answer follows the supplied policy.

Completeness - whether all important required points are addressed.

Tone - whether the response is professional, clear, and concise.

Non-fabrication - whether the response avoids unsupported facts, promises, procedures, or outcomes.

The system returns a structured customer-facing reply, an action field, and an `escalation_needed` flag.

2. Golden set and labelling

The golden set contains **160 unique items** across 10 customer-support categories: **120 development** items and **40 held-out test** items. Each item contains a policy, customer message, and a reference checklist with required points, forbidden claims, the expected decision, and escalation behavior.

Items kept: 160/160 = **100.0%**

Items dropped: 0/160 = **0.0%**

Unique inputs: 160/160

Exact four-score agreement: 153/160 = **95.6%**

Agreement within one point: 160/160 = **100.0%**

Pass/fail agreement: 160/160 = **100.0%**

3. Harness

`runner.py` performs one system-model call per item and stores one JSONL result row containing the input, structured system output, latency, token usage, prompt version, and any error. The model is constrained using provider structured output with the `SupportResponse` schema rather than relying on free-text parsing.

`judge.py` applies the same four-dimensional rubric and is constrained to a structured `JudgeResult`. Results are saved after each request so interrupted or rate-limited runs can resume.

Dataset validation confirmed 160 items, a 120/40 dev-test split, 16 items in each of 10 categories, 160 unique inputs, and **dataset check: PASS**.

4. Judge reliability and bias

Exact judge/consensus agreement: 135/160 = **84.4%**

Within-one agreement: 151/160 = **94.4%**

Pass/fail agreement: 159/160 = **99.4%**

The judge is therefore highly reliable for pass/fail classification, while exact dimensional scoring is less consistent.

Known judge/evaluation disagreements

Item 42: an unsupported carrier-investigation condition affected the non-fabrication score.

Item 45: the response stated a stronger restriction than the policy supported.

Item 61: the response added an unsupported claim that cancellation would not erase the customer's account.

Bias checks

Position-bias verdict changed: 4/14 = **28.6%**.

Verbosity-bias grade increased after padding: 0/14 = **0.0%**.

The position result shows that answer ordering can influence the automated evaluator, while irrelevant padding did not improve scores in the tested cases.

5. System and prompt history

v1 was the baseline prompt. v2 strengthened the prompt by targeting omissions, unsupported promises, review-vs-guarantee mistakes, next-step quality, and consistency between the reply, action, and escalation fields.

System	Split	Pass rate	Average
v1	Dev	117/120 = 97.5%	7.57/8
v2	Dev	120/120 = 100.0%	7.93/8
v2	Held-out test	39/40 = 97.5%	7.85/8

On the development set, v2 improved the pass rate from **97.5%** to **100.0%** and the average score from **7.57/8** to **7.93/8**. The final held-out test result remained strong at **97.5%** with an average of **7.85/8**.

6. Cost and latency

Component	Per item	Per 1,000	Per 100,000
v2 system	\$0.000308	\$0.3075	\$30.75
v2 judge	\$0.000535	\$0.5351	\$53.51

Latency sample size: n = 160

Mean: 1268.26 ms (1.27 s)

p50: 1199.7 ms (1.20 s)

p95: 1673.1 ms (1.67 s)

Minimum: 959.8 ms

Maximum: 2168 ms (2.17 s)

7. Final results

Measure	Result
v1 dev pass rate	117/120 = 97.5%
v1 dev average	7.57/8
v2 dev pass rate	120/120 = 100.0%
v2 dev average	7.93/8
v2 held-out test pass rate	39/40 = 97.5%
v2 held-out test average	7.85/8
Human exact agreement	153/160 = 95.6%
Human within-one agreement	160/160 = 100.0%
Human pass/fail agreement	160/160 = 100.0%
Judge/consensus exact agreement	135/160 = 84.4%
Judge/consensus within-one agreement	151/160 = 94.4%
Judge/consensus pass/fail agreement	159/160 = 99.4%
Position-bias verdict changes	4/14 = 28.6%
Verbosity-bias grade increases	0/14 = 0.0%

Overall, v2 improved over the baseline on the development set and maintained a **97.5% pass rate** on the held-out test set. Human agreement was high, and judge pass/fail agreement with consensus reached **99.4%**. The bias checks show that automated judge scores should still be interpreted with caution, particularly because of position sensitivity.